

GUI-Based Hands-On Lab: Using AWS Secrets Manager with MySQL RDS, step-by-step via the AWS Management Console.

Objective:

- Create a MySQL database using Amazon RDS.
 - Store credentials securely in AWS Secrets Manager.
 - Retrieve and use those credentials from an EC2 instance.
 - (Optional) Enable automatic rotation.
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Lab Architecture Overview:

[Secrets Manager] → securely stores DB credentials
↓
[EC2 Instance] → retrieves secret → connects to
↓
[RDS MySQL DB] → database instance

STEP 1: Create an RDS MySQL Database

1. Go to **AWS Console > RDS > Databases > Create Database**
2. **Engine options:**
 - Choose **MySQL**
3. **Templates:**
 - Select **Free tier** or **Dev/Test**
4. **Settings:**

- DB instance identifier: `mysql-lab-db`
 - Master username: `admin`
 - Master password: `MyStrongPass@123` (store this for now)
5. **DB instance size:** `db.t3.micro` (for free tier)
 6. **Connectivity:**
 - Select existing VPC (or default)
 - Enable **Public access** ☒ (for lab purpose)
 - Select default security group for now
 7. Leave other options default, and click **Create database**
 8. Wait for the DB status to become **Available**
 9. Copy the **RDS endpoint** (needed later)
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STEP 2: Store MySQL Credentials in Secrets Manager

1. Go to **AWS Console > Secrets Manager > Store a new secret**
2. Choose **Credentials for RDS database**
 - Username: `admin`
 - Password: `MyStrongPass@123`
3. Under **Select RDS DB**, choose your MySQL DB instance (`mysql-lab-db`)
4. Click **Next**
5. Name the secret:
`mysql/lab/db-credentials`

6. Description (optional): *My lab MySQL secret*
 7. Leave encryption key as default (AWS managed KMS)
 8. Click **Next > Store**
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✓ STEP 3: Launch EC2 Instance

1. Go to **EC2 > Instances > Launch Instance**
2. **Name:** `mysql-client-ec2`
3. **AMI:** Choose **Amazon Linux 2** or **Ubuntu**
4. **Instance type:** `t2.micro`
5. **Key pair:** Choose/create a key pair for SSH
6. **Network settings:**
 - Create or use a security group with:
 - **Outbound:** Allow all
 - **Inbound:** SSH (port 22), MySQL (port 3306) from your IP
7. Under **Advanced details > IAM role**, attach a role with this policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "secretsmanager:GetSecretValue",
      "Resource": "*"
    }
  ]
}
```

 Tip: Create this IAM role in **IAM > Roles > Create role > EC2** beforehand.

8. Launch the instance.

STEP 4: Connect to EC2 & Test Secret Access

1. Use EC2 Connect or SSH:

```
ssh -i key.pem ec2-user@<EC2 Public IP>
```

2. Install **MySQL client**:

```
sudo yum install -y mysql
```

3. Install AWS CLI (Amazon Linux should already have it):

```
aws sts get-caller-identity
```

(confirm role is attached)

4. Retrieve secret using AWS CLI:

```
aws secretsmanager get-secret-value \  
  --secret-id mysql/lab/db-credentials \  
  --query SecretString \  
  --output text
```

Copy the **username**, **password**, and **host** from the output.

5. Connect to MySQL:

```
mysql -h <RDS-endpoint> -u admin -p
```

Paste the password when prompted.

✓ **STEP 5 (Optional): Enable Secret Rotation**

1. Go to **Secrets Manager** > **mysql/lab/db-credentials**
 2. Click **Edit rotation**
 3. Enable rotation ✓
 4. Choose **Create a new Lambda function**
 5. AWS will create a Lambda function that rotates the password in RDS and Secrets Manager
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📸 **Deliverables for Lab Validation**

- ✓ Screenshot of RDS DB instance in "Available" state
 - ✓ Screenshot of secret in Secrets Manager
 - ✓ Screenshot of EC2 instance running
 - ✓ Screenshot of successful MySQL login from EC2 using retrieved secret
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🧠 **Summary**

Step	Outcome
🎯 RDS DB	MySQL database is created
🔑 Secret	Credentials stored securely
🧠 EC2	Secure retrieval using IAM

 Connect Passwords not hardcoded!

 (Optional) Password rotates automatically via
Lambda
